LITE LED head: 66 x Osram SFH 4545 (950nm, +/−5deg) — 33 strings of 2 on the 4 x AA NiMH pack

I = (VBAT − 2 x 1.5V) / 24R: 100mA on fresh cells (5.4V), 75−83mA for most of the discharge. ~3A peak, ~0.9A average while sending; switched at 38/40kHz by the control board (one low−side switch on LED_RTN; the supply itself is only on while ARM is high).

14 x 5 grid, corners empty (M3 holes): rows of 6 / 7 / 7 / 7 / 6 strings, none crosses rows. Resistors on the back.

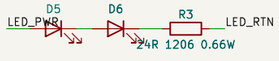
String 1: row 1



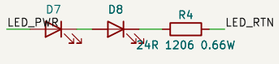
String 2: row 1



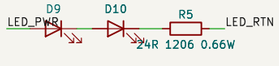
String 3: row 1



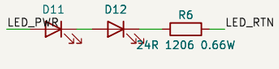
String 4: row 1



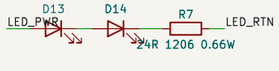
String 5: row 1



String 6: row 1



String 7: row 2



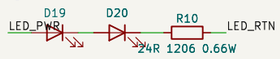
String 8: row 2



String 9: row 2



String 10: row 2



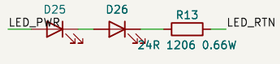
String 11: row 2



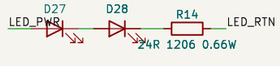
String 12: row 2



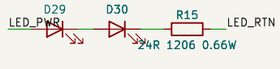
String 13: row 2



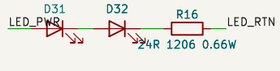
String 14: row 3



String 15: row 3



String 16: row 3



String 17: row 3



String 18: row 3



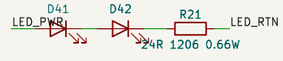
String 19: row 3



String 20: row 3



String 21: row 4



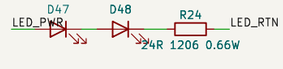
String 22: row 4



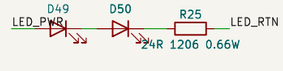
String 23: row 4



String 24: row 4



String 25: row 4



String 26: row 4



String 27: row 4



String 28: row 5



String 29: row 5



String 30: row 5



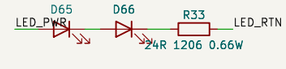
String 31: row 5



String 32: row 5

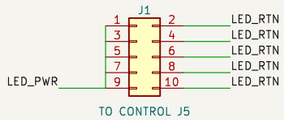


String 33: row 5

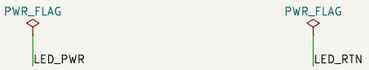


J1 — 2x5 0.1in pin header to control board J5 (pin n = pin n)

Odd pins = LED_PWR, even pins = LED_RTN. Fitted on the BACK of the board.







66 x SFH 4545 950nm, 33 strings of 2, 24R, from the 4 x AA NiMH pack, 99.8 x 44.8 mm

Luke Jamieson

Sheet: /

File: led_head_lite.kicad_sch

Title: Long−range remote LITE — LED head (4 x AA NiMH)

Size: A3	Date: 2026−10−03	**Rev: E**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 1/1